PACMAN Hijinks Firmware and Software (2.X)

Branches, releases and bitwise reproduction

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2026-10-31

1 Purpose and scope

Hijinks (2.X) is the firmware and server written in 2024 for the 10-tile PACMAN 1v5 test stands at LBNL and UC Davis, between the legacy firmware (1.X, companion note) and zipline (3.X). No 2.X firmware was committed in 2024: it lived only in Vivado projects. The record is therefore the project archives, the released bundles, and two branches of `mulhearn/pacman` rebuilt from them. This note describes the branches, the releases, and how each released artifact is reproduced from the repository. Bitstreams are compared by the SHA-256 of their configuration body (first eight hex digits quoted).

2 Branches

- `working-on-hijinks-firmware` (da3f75c): firmware only, linear, on the SYNC FIX build step of the hw1v5 legacy line (3155d4a), so every commit builds by construction. Each 2.X version is one commit restored from its project archive, with the SYNC FIX build fixes applied to the archive's exported block design the same way.
- `working-on-hijinks-software` (af73a4b): the 2024 pacman-server history as committed (server 2.1 to 2.7, 2024-10-02 to 11-08), unchanged, linear.

Both will be superseded by a single hijinks branch of release bundles (under discussion) and then deleted.

2.1 Firmware versions

Version	Commit	Date	Body	Content
—	402413b			firmware only: removes the PetaLinux project and server (now on the software branch), `setup.sh`, and the bare-metal apps needing an AXI GPIO the design lacks
2.1	b163935	2024-10-05	f87eb690	10 tiles, renamed I/O ports, version registers `0x0`–`0xC`, 10-bit masks
2.2	d70db3c	2024-10-07	1431d8cb	hijinks constraints, 32 UARTs across 10 tiles
2.7	96d41a0	2024-10-16	6ba3a82a	UART TX inter-word pause, RX mask (2.3–2.6 were unreleased test increments)
2.8	4b463da	2024-10-26	c4a85eb2	sync/reset on the trigger line too, 32-bit RX mask, 160 µs TX gap
2.9	e250cfd	2024-10-29	0be6948d	sync reset pulse 15 cycles (`C_SYNC_RST_CYCLES` 10 → 15)
2.10	da3f75c	2026-10-21	ede22a64	adjustable MCLK (v1.3-1.0's) on 2.8, for CERN

Table 1: The firmware commits. 2.1–2.9 were rebuilt from the repository (Vivado 2020.2, `recreate_xpr.tcl` then `build_xpr.tcl`, about 90 minutes each) and are bitwise identical to their references (2026-10-13); the commits have since been replayed with the same trees. 2.10 was built from this tree when released.

References: 2.1 the UC Davis bundle 20241005C; 2.2 the bundles 20241007B and 20241010A (two independent bundles, the same bitstream); 2.7 its project archive's own build (no bundle); 2.8 the LBNL bundle 20241026A; 2.9 the LBNL bundle 20241029A; 2.10 its release 20261020A. The project archives are `pacman-fw-2v1.xpr.zip` to `pacman-fw-2v9.xpr.zip` (2v1, 2v2, 2v7, 2v8, 2v9).

3 Releases

Bundle	Kind	FW	Server	Notes
UCD 1v5_20241005A_ucd_syncfix	legacy	SYNC FIX	I2C 5.2	the branching-off point for hijinks
UCD 1v5_20241005B_ucd_devhiijinks	development	SYNC FIX	2.1	server only
UCD 1v5_20241005C_ucd_devhiijinks	development	2.1	2.1	
UCD 1v5_20241007A_ucd_devhijinks	development	2.1	2.2	server only
UCD 1v5_20241007B_ucd_hiijinks	released	2.2	2.2	first hijinks bundle (not successfully deployed)
LBNL 1v5_20241010A_hijinks	released	2.2	2.4	first deployed
LBNL 1v5_20241026A_hiijinks	released	2.8	2.4	
LBNL 1v5_20241029A_hijinks	unreleased	2.9	—	experimental; BOOT.BIN and xsa only
server 2v5, 2v7	software only	(2.8)	2.5, 2.7	tarballs; 2.7 is the CERN deployment
LBNL 1v5_20261020A_hijinks	released	2.10	2.7	released privately for CERN

The released directory names keep their "hiijinks" spelling. Every bundle's `image.ub` is the same (b64633c2). The rootfs images were checked by the server binaries they carry (`pacman_cmdserver`, `pacman_dataserver`): each is exactly a released server tarball's.

Server	`pacman_cmdserver`	`pacman_dataserver`	Shipped in
2.1	27f457d0	19b33c8c	tarball 2v1 (= 05B's); rootfs of 05C
2.2	af9df9ab	f0c56599	tarball 2v2; rootfs of 07A, 07B
2.4	629674af	230cd1df	tarball 2v4; rootfs of 10A, 26A
2.5	8e9c70b8	665e4383	tarball 2v5
2.7	61f8cb40	13fcbb44	tarball 2v7

Table 3: The released server binaries (SHA-256, first eight hex digits).

4 Reproducing the server

The server is a PetaLinux 2018.3 application (`petalinux/pkg/pacman-server`, recipe `pacman-server.bb`), built by `petalinux-build -c pacman-server` (`petalinux/scripts/build_pacman_server.sh`, as the branch's README describes) and shipped as the files of the resulting RPM. The binaries are stripped and carry no build dates. A build of af73a4b on felix (2026-10-11) gives binaries of exactly the released 2.7 sizes, with the same sections and the same strings, but in a different order: the Makefile collects its sources with `$(wildcard src/*.cc)`, whose order is the directory order of the build machine's file system, so the objects are linked in a different order. A pilot rebuilds 2.7 with the objects kept, finds the release's link order from its code, and tests whether linking in that order reproduces the release bitwise; the other servers follow the pilot's method.

5 Still open

- The 2.7 server pilot, then 2.1, 2.2, 2.4 and 2.5 (2.2 and 2.5 span several commits: the release commit is to be found).
- Every firmware version built once more from a fresh clone of the repository.
- One hijinks branch of release bundles: one commit per release, each a real merge of the firmware and software it brings in, confirmed by a tree check against the proven trees; annotated tags carrying each build result.
- Validation of 2.10 with the hardware loopback card at 5 and 10 MHz.